Order Continuity for Closed Subspaces of Free p-Convex Banach Lattices: The Complemented Case and a Finite-p Counterexample

Independent research note

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Abstract

We first verify a supplied partial argument: if a bounded operator $T : F \to E$ has a bounded left inverse, then the induced lattice homomorphism $\widehat{T} : \mathrm{FBL}^{(p)}[F] \to \mathrm{FBL}^{(p)}[E]$ is order continuous for every $1 \leq p \leq \infty$. The proof is correct, but this is essentially the complemented-range part of the original proof of Oikhberg–Taylor–Tradacete– Troitsky, the part explicitly retained in their corrigendum.

We then answer the remaining finite-p closed-subspace question negatively. Let F be the closed span of the Rademacher functions in $L_1[0,1]$. Thus F is isomorphic to ℓ_2. For every $1 \leq p < \infty$, the canonical lattice homomorphism

$$\mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[L_1[0,1]]$$

is not order continuous. The same pair $F \subset L_1[0,1]$ works simultaneously for all finite p. Combined with the theorem of Azouzi–Dhifaoui for $p = \infty$, this gives a sharp finite-versus-infinite p dichotomy. The counterexample and proof below are independent and have not been peer reviewed; a literature search through the date above did not locate a prior finite-p resolution.

Packet status

Run `fa_banach_001`.

Source arXiv id

`2508.11648`.

Source paper

Y. Azouzi and W. Dhifaoui, *Order denseness in free Banach lattices.* Source record: `arXiv:2508.11648v1`.

Packet classification

Candidate counterexample, likely valid subject to human mathematical review.

Upgrade note

This packet supersedes the earlier partial packet for the same arXiv id. The earlier packet established only the complemented-range subcase, which the corrigendum to Oikhberg–Taylor–Tradacete–Troitsky already retains. The new contribution recorded here is the finite-p obstruction for an uncomplemented closed subspace.

The source paper frames the finite-p closed-subspace case as the remaining open part after its $p = \infty$ theorem:

This shows that $y = 0$ and consequently, $[0, f] \cap \mathrm{FVL}[E] = \{0\}$ as required. Therefore, we conclude that the free vector lattice $\mathrm{FVL}[E]$ is not order dense in $\mathrm{FBL}^{(p)}[E]$, which completes the proof. ■

Consider two Banach spaces E and F and a bounded operator $T : F \longrightarrow E$. Then as noted above we can associate to T, in a natural way, a lattice homomorphism $\overline{T} : \mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[E]$ by extending T to $\mathrm{FBL}^{(p)}[F]$ and then compose with the natural embedding $E \hookrightarrow \mathrm{FBL}^{(p)}[E]$. One of the main ideas studied in [13] is the relationship between operators T and $\overline{T}$. In [13] the authors showed that if $\mathcal{P}$ is one of the properties – injectivity, surjectivity and bijectivity – then T has property $\mathcal{P}$ iff $\overline{T}$ does. Among these, only the proof of injectivity requires order denseness. For this reason we provide a corrected proof for it.

Proposition 6 *Let E and F be two Banach spaces, $T \in L(F, E)$ and $\overline{T} : \mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[E]$ be the associated lattice homomorphism introduced above. Then T is injective iff $\overline{T}$ is injective.*

Proof. We need only to prove the forward implication. Assume then that T is injective and let $f \in \mathrm{FBL}^{(p)}[F]$ such that $\overline{T}f = 0$. Then f vanishes on the range $T^*(E^*)$ of T^*. As f is bounded weak* continuous f vanishes on $\overline{T^*(E^*)}^{bw^*}$. As T is injective T^* has a weak* dense range and it is sufficient to show that $\overline{T^*(E^*)}^{bw^*} = \overline{T^*(E^*)}^{w^*}$. But this follows from Banach-Dieudonné Theorem (Theorem 2). ■

In [13, Theorem 3.4] it was shown if F is closed subspace of E and $\iota : F \hookrightarrow E$ is the inclusion map then $\overline{\iota} : \mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[E]$ is order continuous. However, the proof relies on order denseness of $\mathrm{FVL}[F]$ in $\mathrm{FBL}^{(p)}[F]$ which holds only in the finite-dimensional. The converse problem was also raised (see [13, Remark 3.5]). What we have been were able to prove that the former result is true for $p = \infty$; for other values of p, the question remains open. Furthermore, we give a partial answer concerning the converse problem. Before presenting the proof of these results let us state an interesting result, which is very close to [13, Lemma 2.9].

Theorem 7 *Let U be a bounded weak* open subset of E^* and $0 \neq x_0^* \in U$. Then there exists f in $\mathrm{FBL}^{(\infty)}[E]^+$ such that $f(x_0^*) \neq 0$ and $f = 0$ outside of $\mathrm{cone}(U)$. Moreover if $h \in H_\infty[E]^+$ satisfies $h(x^*) > 1$ for all $x^* \in U$ then*

1 Background and statement of the answer

All Banach spaces and Banach lattices are real. For a Banach space X and $1 \leq p \leq \infty$, write $\mathrm{FBL}^{(p)}[X]$ for the free p-convex Banach lattice generated by X, with canonical generator map $x \mapsto \delta_x^X$. If $A : X \to Y$ is bounded, its canonical extension is denoted by

$$\widehat{A} : \mathrm{FBL}^{(p)}[X] \longrightarrow \mathrm{FBL}^{(p)}[Y].$$

In the standard functional representation,

$$(\widehat{A}f)(y^*) = f(A^*y^*) \qquad (y^* \in Y^*), \tag{1.1}$$

and the construction is functorial:

$$\widehat{BA} = \widehat{B}\,\widehat{A}. \tag{1.2}$$

Oikhberg, Taylor, Tradacete and Troitsky originally claimed that, whenever F is a closed subspace of E, the inclusion-induced map $\mathrm{FBL}^{(p)}[F] \to \mathrm{FBL}^{(p)}[E]$ is order continuous for every p [4, Theorem 3.4]. The proof used an order-density assertion that fails in infinite dimensions. Azouzi and Dhifaoui proved that the inclusion result remains true for $p = \infty$, and left the finite-p case open [1, Theorem 11]. The subsequent corrigendum says explicitly that the old proof works only for complemented subspaces [5].

For definiteness, throughout the construction we use the Rademacher system

$$r_n(t) = (-1)^{\lfloor 2^{n+1}t \rfloor} \qquad (0 \le t < 1,\ n \ge 0),$$

with arbitrary values at the dyadic endpoints. Thus the r_n's are independent $\{-1,1\}$-valued functions and are pairwise orthogonal in $L_2[0,1]$.

The principal result of this note is the following.

Theorem 1.1. *Let $(r_n)_{n\ge 0}$ be the Rademacher functions on $[0,1]$, let*

$$E = L_1[0,1], \qquad F = \overline{\operatorname{span}}\{r_n : n \ge 0\}^{L_1},$$

and let $\iota : F \hookrightarrow E$ be the inclusion. Then F is a closed subspace isomorphic to ℓ_2, and for every $1 \le p < \infty$ the lattice homomorphism

$$\widehat{\iota} : \mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[E]$$

is not order continuous.

Thus the answer for arbitrary closed subspaces is positive at $p = \infty$ and negative at every finite p.

2 Proof intuition

The counterexample compares the same decreasing free-lattice sequence in two ambient spaces. In $H = \ell_2$, the vectors $u_n = e_0 + s_n e_n$ are chosen with $s_n \downarrow 0$ very slowly on very large coordinate blocks. Finite-p admissible families can concentrate on those blocks strongly enough to force any positive common lower bound of

$$f_n = \bigwedge_{k=0}^{n} |\delta^H_{u_k}|$$

to vanish.

After the Rademacher embedding $H \hookrightarrow L_1[0,1]$, the same perturbation directions behave differently. Finite joins of $|\delta^{L_1}_{r_k}|$ have uniformly bounded $\mathrm{FBL}^{(p)}[L_1]$-norm, so the images of the f_n's form a norm-Cauchy decreasing sequence with a nonzero limit. A positive operator that sends a decreasing-to-zero sequence to one with a nonzero lower bound cannot be order continuous.

3 Finite-p preliminaries

Fix $1 \le p < \infty$. We use the concrete norm formula for $\mathrm{FBL}^{(p)}[X]$: if f is a positively homogeneous function on X^*, then

$$\|f\|_{\mathrm{FBL}^{(p)}[X]} = \sup\left\{ \left(\sum_{j=1}^{m} |f(x_j^*)|^p\right)^{1/p} : \sup_{x\in B_X} \left(\sum_{j=1}^{m} |x_j^*(x)|^p\right)^{1/p} \le 1 \right\}. \tag{1}$$

The free lattice $\mathrm{FBL}^{(p)}[X]$ is the norm-closed sublattice generated by $\{\delta_x^X : x \in X\}$, where $\delta_x^X(x^*) = x^*(x)$. In particular,

$$|f(x^*)| \le \|f\|_{\mathrm{FBL}^{(p)}[X]} \|x^*\| \qquad (f \in \mathrm{FBL}^{(p)}[X],\ x^* \in X^*). \tag{2}$$

Indeed, for $x^* \neq 0$, the one-element family $x^*/\|x^*\|$ is admissible in (1).

We shall repeatedly use the following elementary order criterion. If (x_α) is a decreasing net in a vector lattice with $x_\alpha \ge 0$, then $x_\alpha \downarrow 0$ exactly when there is no nonzero positive common lower bound. Consequently, to disprove order continuity of a positive operator it suffices to find $x_n \downarrow 0$ whose images have a nonzero positive common lower bound.

4 Verification of the complemented-range argument

The supplied partial proof is correct. We record it in operator form because it also fixes the orientation of all functorial identities.

Theorem 4.1. *Let $1 \le p \le \infty$, and let $T : F \to E$ be bounded. Suppose that there is a bounded operator $R : E \to F$ such that $RT = \mathrm{Id}_F$. Then*

$$\widehat{T} : \mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[E]$$

is order continuous.

Proof. Let $f_\alpha \downarrow 0$ in $\mathrm{FBL}^{(p)}[F]_+$, and suppose that $0 \le h \le \widehat{T} f_\alpha$ for every α, where $h \in \mathrm{FBL}^{(p)}[E]$. We prove $h = 0$.

Assume otherwise and choose $x^* \in E^*$ with $h(x^*) > 0$. Put $y^* = T^* x^*$. Then $y^* \neq 0$, since otherwise

$$0 < h(x^*) \le (\widehat{T} f_\alpha)(x^*) = f_\alpha(0) = 0.$$

Set $d = x^* - R^* y^*$. Since $RT = \mathrm{Id}_F$, one has $d(Tu) = 0$ for every $u \in F$. Choose $u_0 \in F$ such that $y^*(u_0) = 1$, and define

$$Se = e + d(e) T u_0 \qquad (e \in E).$$

Then S is bounded, $ST = T$, and

$$S^* R^* y^* = x^*.$$

Indeed, for $e \in E$,

$$(S^* R^* y^*)(e) = y^*(Re + d(e) RT u_0) = (R^* y^*)(e) + d(e) = x^*(e).$$

By functoriality,

$$\widehat{S}\,\widehat{T} = \widehat{ST} = \widehat{T}, \qquad \widehat{R}\,\widehat{T} = \widehat{RT} = \mathrm{Id}_{\mathrm{FBL}^{(p)}[F]}.$$

Therefore

$$0 \le \widehat{R}\,\widehat{S}h \le \widehat{R}\,\widehat{S}\,\widehat{T}f_\alpha = \widehat{R}\,\widehat{T}f_\alpha = f_\alpha$$

for every α. But

$$(\widehat{R}\,\widehat{S}h)(y^*) = h(S^*R^*y^*) = h(x^*) > 0,$$

so $\widehat{R}\,\widehat{S}h$ is a nonzero positive common lower bound for (f_α), a contradiction. $\square$

Remark 4.2. The hypothesis in Theorem 4.1 says precisely that T is an isomorphic embedding with complemented range. Factoring T through its range shows that this is the complemented-subspace case of the original theorem. The original paper already proved that case, and its corrigendum explicitly states that this is the part of the proof that remains valid. Thus the supplied argument is mathematically sound, but it is not a new subcase of the literature [5].

5 A decreasing source sequence with infimum zero

Let

$$H = \ell_2(\mathbb{N}_0), \qquad \mathbb{N}_0 = \{0, 1, 2, \ldots\},$$

with unit vector basis $(e_n)_{n\ge 0}$. We first choose a single slowly decaying sequence that will work for every finite p.

For $j \ge 1$, set

$$m_j = 2^{j^2}, \qquad M_j = \sum_{i=1}^{j} m_i, \qquad I_j = \{M_{j-1}+1, \ldots, M_j\},$$

where $M_0 = 0$. Define

$$s_n = \frac{1}{j} \qquad (n \in I_j). \tag{3}$$

Then (s_n) is non-increasing and tends to zero. Moreover, for every finite q,

$$\frac{1}{j}\, m_j^{1/q} \longrightarrow \infty. \tag{4}$$

Set

$$u_0 = e_0, \qquad u_n = e_0 + s_n e_n \quad (n \ge 1), \qquad f_n = \bigwedge_{k=0}^{n} |\delta^H_{u_k}| \in \mathrm{FBL}^{(p)}[H]_+. \tag{5}$$

We need one density observation.

Lemma 5.1. *The union of the vector sublattices generated by $\delta^H_{e_0}, \ldots, \delta^H_{e_N}$, $N \ge 0$, is norm dense in* $\mathrm{FBL}^{(p)}[H]$.

Proof. The finitely supported vectors are dense in H, and $x \mapsto \delta^H_x$ is an isometry. Hence the closed sublattice generated by $\{\delta^H_x : x \in c_{00}\}$ contains every δ^H_x, $x \in H$, and is therefore all of $\mathrm{FBL}^{(p)}[H]$. Every lattice expression involving finitely many finitely supported vectors is contained in the sublattice generated by $e_0, \ldots, e_N$ for some N. $\square$

Proposition 5.2. *For every $1 \le p < \infty$, the sequence in* (5) *satisfies*

$$f_n \downarrow 0 \quad \textit{in } \mathrm{FBL}^{(p)}[H].$$

Proof. The sequence is decreasing and positive. Suppose that $0 \le g \le f_n$ for every n. We prove $g = 0$.

Assume $g \neq 0$. By positive homogeneity and (2), choose $x^* \in H^*$ with $\|x^*\| = 1$ and

$$\eta := g(x^*) > 0.$$

Write

$$a = x^*(e_0), \qquad b_n = x^*(e_n) \quad (n \ge 1).$$

Since $g \le f_0 = |\delta_{e_0}^H|$, we have $a \neq 0$. By Lemma 5.1, choose N and an element ϕ of the sublattice generated by $\delta_{e_0}^H, \dots, \delta_{e_N}^H$ such that

$$\|g - \phi\|_{\mathrm{FBL}^{(p)}[H]} < \frac{\eta}{4}. \tag{6}$$

Then (2) gives

$$\phi(x^*) > \frac{3\eta}{4}. \tag{7}$$

For $n > N$, define

$$d_n = b_n + \frac{a}{s_n}, \qquad y_n^* = x^* - d_n e_n^*.$$

Here e_n^* is the n-th coordinate functional. We have

$$y_n^*(u_n) = a + s_n(b_n - d_n) = 0.$$

Thus

$$g(y_n^*) = 0, \qquad \phi(y_n^*) = \phi(x^*) \quad (n > N), \tag{8}$$

because $g \le f_n$, while ϕ depends only on the first $N+1$ coordinates.

Put

$$w_n = \frac{1}{|d_n|} = \frac{s_n}{|a + s_n b_n|}.$$

Since $b_n \to 0$, after increasing N we may assume that

$$c s_n \le w_n \le C s_n \qquad (n > N) \tag{9}$$

for constants $c, C > 0$ independent of n.

Choose a block $I_j \subset \{n > N\}$. We now select nonnegative weights $(\lambda_n)_{n \in I_j}$. If $p \ge 2$, set

$$\lambda_n = w_n.$$

Then $|\lambda_n d_n| = 1$, and for every $x = (x_k) \in H$,

$$\left(\sum_{n \in I_j} |\lambda_n d_n x_n|^p \right)^{1/p} \le \left(\sum_{n \in I_j} |x_n|^2 \right)^{1/2} \le \|x\|. \tag{10}$$

Moreover, by (9) and (4),

$$L_j := \left(\sum_{n \in I_j} \lambda_n^p \right)^{1/p} \ge \frac{c}{j} m_j^{1/p} \longrightarrow \infty.$$

If $1 \le p < 2$, set
$$\alpha_j = m_j^{1/2-1/p}, \qquad \lambda_n = \alpha_j w_n \quad (n \in I_j).$$
Since $\|z\|_{\ell_p^{m_j}} \le m_j^{1/p-1/2}\|z\|_{\ell_2^{m_j}}$, we again have
$$\left(\sum_{n\in I_j} |\lambda_n d_n x_n|^p\right)^{1/p} = \alpha_j \left(\sum_{n\in I_j} |x_n|^p\right)^{1/p} \le \|x\|. \tag{11}$$
This time
$$L_j := \left(\sum_{n\in I_j} \lambda_n^p\right)^{1/p} \ge \frac{c}{j}\, m_j^{1/2} \longrightarrow \infty.$$
Thus, in both cases, $L_j \to \infty$ and the diagonal estimate
$$\left(\sum_{n\in I_j} |\lambda_n d_n x_n|^p\right)^{1/p} \le \|x\| \tag{12}$$
holds.

For $x \in H$, Minkowski's inequality and $\|x^*\| = 1$ now give
$$\begin{aligned}\left(\sum_{n\in I_j} |\lambda_n y_n^*(x)|^p\right)^{1/p} &\le L_j|x^*(x)| + \left(\sum_{n\in I_j} |\lambda_n d_n x_n|^p\right) \\ &\le (L_j+1)\|x\|.\end{aligned}$$
Hence the finite family
$$\left(\frac{\lambda_n}{L_j+1} y_n^*\right)_{n\in I_j}$$
is admissible in (1). Using (8) and positive homogeneity, we obtain
$$\begin{aligned}\|g-\phi\|_{\mathrm{FBL}^{(p)}[H]} &\ge \left(\sum_{n\in I_j} |(g-\phi)\left(\frac{\lambda_n}{L_j+1} y_n^*\right)|^p\right)^{1/p} \\ &= \phi(x^*)\frac{L_j}{L_j+1}.\end{aligned}$$
Letting $j \to \infty$, the right-hand side tends to $\phi(x^*)$, which is larger than $3\eta/4$ by (7). This contradicts (6). Therefore $g = 0$, and $f_n \downarrow 0$. $\square$

6 The images acquire a nonzero lower bound in L_1

We first record the ambient norm estimate that makes the construction work.

Lemma 6.1. *Let $J \subset \mathbb{N}_0$ be finite and nonempty. For every $1 \le p < \infty$,*
$$\|\bigvee_{k\in J} |\delta_{r_k}^{L_1}|\|_{\mathrm{FBL}^{(p)}[L_1[0,1]]} \le 1.$$

Proof. Let $\varphi_1, \dots, \varphi_m \in L_\infty[0,1]$ be admissible in (1); equivalently, the operator

$$A : L_1[0,1] \longrightarrow \ell_p^m, \qquad Ax = \left(\int_0^1 x(t)\varphi_j(t)\, dt \right)_{j=1}^m,$$

has norm at most one. Its adjoint has norm at most one, and finite-dimensional duality therefore gives

$$\left(\sum_{j=1}^m |\varphi_j(t)|^p \right)^{1/p} \le 1 \quad \text{for almost every } t. \tag{13}$$

For completeness, one obtains a common null set by testing A^* on a countable dense subset of the unit ball of the conjugate finite-dimensional space.

Since $|r_k| = 1$ almost everywhere, Minkowski's integral inequality and (13) yield

$$\begin{aligned} &\left(\sum_{j=1}^m \left(\max_{k \in J} \left| \int_0^1 r_k(t)\varphi_j(t)\, dt \right| \right)^p \right)^{1/p} \\ &\le \left(\sum_{j=1}^m \left(\int_0^1 |\varphi_j(t)|\, dt \right)^p \right)^{1/p} \\ &\le \int_0^1 \left(\sum_{j=1}^m |\varphi_j(t)|^p \right)^{1/p} dt \le 1. \end{aligned}$$

Taking the supremum over admissible families proves the claim. $\square$

Let

$$T : H \longrightarrow L_1[0,1], \qquad T(x_0, x_1, \dots) = \sum_{n=0}^\infty x_n r_n.$$

Khintchine's inequality implies that T is an isomorphic embedding. Consequently,

$$F = T(H) = \overline{\operatorname{span}}\{r_n : n \ge 0\}^{L_1}$$

is closed and isomorphic to ℓ_2. For the vectors from (5), put

$$v_0 = Tu_0 = r_0, \qquad v_n = Tu_n = r_0 + s_n r_n \quad (n \ge 1).$$

For fixed finite p, define

$$G_n = \bigwedge_{k=0}^n |\delta_{v_k}^{L_1}| \in \mathrm{FBL}^{(p)}[L_1[0,1]]_+. \tag{14}$$

By functoriality,

$$G_n = \widehat{T} f_n. \tag{15}$$

Proposition 6.2. *For every $1 \le p < \infty$, the sequence (G_n) converges in norm to an element $h_p \in \mathrm{FBL}^{(p)}[L_1[0,1]]_+$ such that*

$$0 < h_p \le G_n \qquad (n \ge 0).$$

More precisely,

$$\|G_n - G_m\|_{\mathrm{FBL}^{(p)}[L_1]} \le s_{n+1} \qquad (m > n).$$

Proof. For $m > n$, the scalar lattice identity

$$a - (a \wedge b_{n+1} \wedge \cdots \wedge b_m) = \bigvee_{k=n+1}^{m} (a - b_k)^+$$

applied pointwise gives

$$G_n - G_m = \bigvee_{k=n+1}^{m} \left(G_n - |\delta_{v_k}^{L_1}| \right)^+ .$$

Since $G_n \leq |\delta_{v_0}^{L_1}|$, the standard lattice inequality $||x| - |y|| \leq |x - y|$ yields

$$\begin{aligned} 0 \leq G_n - G_m &\leq \bigvee_{k=n+1}^{m} ||\delta_{v_0}^{L_1}| - |\delta_{v_k}^{L_1}|| \\ &\leq \bigvee_{k=n+1}^{m} |\delta_{v_0 - v_k}^{L_1}| \\ &= \bigvee_{k=n+1}^{m} s_k |\delta_{r_k}^{L_1}| \\ &\leq s_{n+1} \bigvee_{k=n+1}^{m} |\delta_{r_k}^{L_1}|. \end{aligned}$$

Lemma 6.1 proves the norm estimate. Thus (G_n) is norm Cauchy; let its limit be h_p. The positive cone is closed and $G_m \leq G_n$ for $m > n$, so $0 \leq h_p \leq G_n$ for every n.

Finally, regard r_0 as an element of $L_\infty = L_1^*$. Pairwise orthogonality of the Rademacher functions in L_2 gives

$$\langle r_0, v_0 \rangle = 1, \qquad \langle r_0, v_k \rangle = 1 + s_k \int_0^1 r_0 r_k = 1 \quad (k \geq 1).$$

Hence $G_n(r_0) = 1$ for every n. Evaluation at r_0 is continuous by (2), so $h_p(r_0) = 1$, and therefore $h_p > 0$. $\square$

7 Proof of the counterexample and the sharp dichotomy

Proof of Theorem 1.1. Let $U : H \to F$ be the operator T viewed as a map onto its range. Since U is an isomorphism of Banach spaces, functoriality shows that

$$\widehat{U} : \mathrm{FBL}^{(p)}[H] \longrightarrow \mathrm{FBL}^{(p)}[F]$$

is a lattice isomorphism, with inverse $\widehat{U^{-1}}$. Set

$$g_n = \widehat{U} f_n \in \mathrm{FBL}^{(p)}[F].$$

Proposition 5.2 gives $g_n \downarrow 0$. Since $T = \iota U$, equations (15) and (14) give

$$\widehat{\iota} g_n = \widehat{\iota} \widehat{U} f_n = \widehat{T} f_n = G_n.$$

By Proposition 6.2, the sequence (G_n) has the nonzero positive common lower bound h_p. Therefore $\widehat{\iota}$ is not order continuous.

The spaces E and F, and the sequence (s_n), were chosen independently of p. Hence the same closed Hilbertian subspace $F \subset L_1[0,1]$ works simultaneously for every finite p. $\square$

Corollary 7.1 (Sharp finite-versus-infinite dichotomy). *For the Rademacher subspace $F \subset L_1[0,1]$, the inclusion-induced map*

$$\mathrm{FBL}^{(p)}[F] \longrightarrow \mathrm{FBL}^{(p)}[L_1[0,1]]$$

is not order continuous for $1 \leq p < \infty$, whereas it is order continuous for $p = \infty$.

Proof. The finite-p assertion is Theorem 1.1. The $p = \infty$ assertion is the closed-range embedding theorem of Azouzi–Dhifaoui [1, Theorem 11]. □

Remark 7.2. The mechanism is worth emphasizing. In the source Hilbert space, the perturbations $s_n e_n$ are distributed over very large coordinate blocks. Finite-p admissible families can amplify those blocks enough to force the only common lower bound of (f_n) to be zero. After the Rademacher embedding into L_1, finite joins of the perturbation directions have uniformly bounded free-lattice norm. The image sequence therefore converges in norm to a nonzero lower bound. This mismatch is exactly the failure of regularity.

8 Literature status

The relevant chronology is as follows.

- Oikhberg–Taylor–Tradacete–Troitsky, Theorem 3.4, asserted order continuity for every closed subspace and every p. Its complemented subspace argument is valid [4, 5].
- Azouzi–Dhifaoui identified the order-density gap, proved the full closed-range embedding result for $p = \infty$, and stated that the other values of p remained open [1].
- The corrigendum of Oikhberg–Taylor–Tradacete–Troitsky, published on June 19, 2026, says that the proof of Theorem 3.4 works only when the subspace is complemented and that the claim had not been used elsewhere in the literature [5].

Exact-title, theorem-number, citation, and arXiv searches through June 22, 2026 did not locate a published or posted finite-p proof or counterexample. Accordingly, the finite-p construction above should be treated as an independent, unrefereed argument rather than as a certified novelty claim.

References

[1] Y. Azouzi and W. Dhifaoui, *Order denseness in free Banach lattices*, arXiv:2508.11648v1, 2025.

[2] A. Avilés, J. Rodríguez, and P. Tradacete, *The free Banach lattice generated by a Banach space*, J. Funct. Anal. **274** (2018), no. 10, 2955–2977.

[3] H. Jardón-Sánchez, N. J. Laustsen, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices under convexity conditions*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM **116** (2022), article 15.

[4] T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Free Banach lattices*, J. Eur. Math. Soc. **28** (2026), no. 10, 4387–4514; DOI 10.4171/JEMS/1552.

[5] T. Oikhberg, M. A. Taylor, P. Tradacete, and V. G. Troitsky, *Corrigendum to "Free Banach lattices"*, J. Eur. Math. Soc. **28** (2026), no. 10, 4515–4518; DOI 10.4171/JEMS/1802.